Public Health 255 (2026) 106272

Contents lists available at ScienceDirect

Public Health

journal homepage: www.elsevier.com/locate/puhe

Review Article

Optimizing exercise prescription parameters for ADHD in children: A systematic review and meta-analysis

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ARTICLE INFO

Keywords:
Physical exercise
ADHD children
Exercise prescription
Intervention effects
Meta-analysis

ABSTRACT

Objectives: This study aimed to systematically evaluate the effects of physical exercise on children with Attention Deficit Hyperactivity Disorder (ADHD) and provide evidence for optimal exercise prescription.
Study design: Systematic review and meta-analysis of randomized controlled trials following PRISMA guidelines.
Methods: We searched databases including PubMed, Web of Science, Cochrane Library, ScienceDirect, and EBSCO from inception to July 20, 2025, incorporating 20 randomized controlled trials with 48 effect sizes. We conducted meta-analysis and subgroup analysis using a random-effects model.
Results: Physical exercise may yield modest improvements in ADHD symptoms (SMD = 0.27, 95% CI [0.00-0.54]). Therapeutic and alternative exercise modalities (SMD = 0.45, 95% CI [0.26-0.65]) showed greater effects than Aerobic and Endurance Training (SMD = 0.16, 95% CI [0.00-0.32]). Mid-term interventions (12-24 weeks, SMD = 0.28, 95% CI [0.11-0.46]) outperformed short-term programs. Mid-frequency (twice weekly, SMD = 0.52, 95% CI [0.33-0.71]) and long duration sessions (60-90 min, SMD = 1.01, 95% CI [0.71-1.31]) demonstrated optimal effectiveness.
Conclusions: Physical exercise serves as a potentially beneficial adjunctive treatment for improving ADHD symptoms in children. Exploratory subgroup analyses suggest that therapeutic and alternative exercise modalities lasting 12-24 weeks, performed twice weekly for 60-90 min per session, may be associated with larger effect sizes, though these findings require further confirmation. These findings provide important references for clinical practice in managing ADHD in children.

1. Introduction

Attention Deficit Hyperactivity Disorder (ADHD), a common neurodevelopmental disorder, has become a global public health concern. ADHD symptoms, including inattention, hyperactivity, and impulsivity, typically emerge before school age and persist for more than six months, significantly impacting children's development.[1] The World Health Organization reports that ADHD affects approximately 2.9% of children worldwide,[2] with a rising prevalence in recent years.[3] These children often struggle with concentration, leading to deficits in language communication,[4] literacy,[5] and overall physical and mental health,[6] which increase the risk of academic failure, social difficulties, and emotional disorders. ADHD imposes substantial economic burdens, with annual costs exceeding $1000-$2000 per child in the United States[7] and €5000-€10,000 in Europe.[8] Although pharmacotherapy remains the most common treatment, it often causes side effects such as headaches and stomach pain, limiting its use.[9]

Given pharmacotherapy limitations, researchers have explored alternative treatments, with physical exercise emerging as a promising non-pharmacological intervention. Meta-analyses have systematically evaluated physical exercise effects on ADHD symptoms,[10–12] demonstrating improvements in attention and behavioral problems,[10] positive effects on functional outcomes,[11] and effective symptom alleviation, particularly for executive functions.[12] However, debate persists regarding appropriate exercise prescription parameters, with conflicting recommendations on exercise types, intensities, and durations.[13–15] While substantial evidence indicates that physical exercise significantly improves attention, executive function, and behavioral cognition in children with ADHD,[16,17] inconsistent findings have also been reported,[18] underscoring the need for evidence-based exercise prescription guidelines. Existing meta-analyses have limitations including small sample sizes, high heterogeneity among studies, and lack of systematic

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https://doi.org/10.1016/j.puhe.2026.106272
Received 21 October 2025; Received in revised form 17 February 2026; Accepted 24 March 2026
Available online 2 April 2026
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analysis of exercise parameter relationships with symptom improvement, hindering clinical translation.

This study aims to fill this gap by conducting a comprehensive meta-analysis of physical exercise effects on core ADHD symptoms in children, focusing on identifying effective exercise prescription parameters. By including high-quality RCTs and systematically evaluating different exercise types, frequencies, durations, and intensities, this study seeks to provide robust evidence for developing evidence-based exercise intervention strategies for children with ADHD.

2. Methods

2.1. Literature search

A literature search was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines[19] (Fig. 1). This meta-analysis was prospectively registered with PROSPERO (registration number: CRD420251013415). PubMed, Web of Science, Cochrane Library, ScienceDirect, and EBSCO databases were searched from inception to July 20, 2025. The English search strategy combined MeSH terms and free-text words: ("Exercises"[Title/Abstract]) OR ("physical activity"[Title/Abstract]) OR ("physical exercises"[Title/Abstract]) OR (sport[Title/Abstract]) AND ("Attention Deficit Disorder with Hyperactivity"[Mesh Terms]) OR ("ADHD"[Title/Abstract]) OR ("Attention Deficit Hyperactivity Disorder"[Title/Abstract]) OR ("Hyperkinetic Disorder"[Title/Abstract]) OR ("Attention Deficit Disorder"[Title/Abstract]) AND ("Child"[Mesh Terms]) OR ("Adolescent"[Mesh Terms]) OR ("child"[Title/Abstract]) OR ("adolescen"[Title/Abstract]) OR ("youth"[Title/Abstract]) OR ("teen"[Title/Abstract]) AND ("Intervention"[Title/Abstract]) OR ("Program"[Title/Abstract]) OR ("Impact"[Title/Abstract]) OR ("Effect"[Title/Abstract]) OR ("Outcome"[Title/Abstract]) AND ("randomized controlled trial"[Publication Type]) OR ("controlled clinical trial"[Publication Type]) OR ("meta-analysis"[Publication Type]).

Studies were selected based on: (1) participants aged 4-18 years; (2) interventions including exercise or physical activity; (3) high-quality randomized controlled trial design. Reference lists of included studies were manually screened for additional literature. Two researchers independently screened studies and extracted data, with disagreements resolved through discussion or arbitration by a third researcher. After screening, study quality was assessed using the Cochrane risk of bias tool and data on study characteristics, interventions, and ADHD symptoms were extracted for meta-analysis.

1."Exercise"[Mesh]

2."Exercises"[Title/Abstract]) OR ("physical activity"[Title/Abstract]) OR ("physical exercises"[Title/Abstract]) OR (sport[Title/Abstract])

3.#1 OR #2

4."Attention Deficit Disorder with Hyperactivity"[Mesh Terms]) OR ("ADHD"[Title/Abstract]) OR ("Attention Deficit Hyperactivity Disorder"[Title/Abstract]) OR ("Hyperkinetic Disorder"[Title/Abstract]) OR ("Attention Deficit Disorder"[Title/Abstract])

5."Child"[Mesh Terms]) OR ("Adolescent"[Mesh Terms]) OR ("child"[Title/Abstract]) OR ("adolescen"[Title/Abstract]) OR ("youth"[Title/Abstract]) OR ("teen"[Title/Abstract])

6."Intervention"[Title/Abstract]) OR ("Program"[Title/Abstract]) OR ("Impact"[Title/Abstract]) OR ("Effect"[Title/Abstract]) OR ("Outcome"[Title/Abstract])

7."randomized controlled trial"[Publication Type]) OR ("controlled clinical trial"[Publication Type]) OR("meta-analysis"[Publication Type])

8.#3 AND #4 AND #5 AND #6 AND #7

Fig. 1. PubMed search strategy.

2.2. Inclusion and exclusion criteria

2.2.1. Inclusion criteria

Studies were included based on the following criteria: (1) Participants were children aged 4-18 years with ADHD symptoms, without cognitive impairment or history of severe physical illness. Children were defined as individuals aged 4-18 years, consistent with definitions used by the United Nations Convention on the Rights of the Child[20] and the American Academy of Pediatrics,[21] encompassing both childhood and adolescence across the developmental spectrum; (2) Interventions involved physical exercise, including aerobic and endurance training, therapeutic and alternative exercises for the experimental group, while the control group received no physical exercise intervention; (3) Study design was a high-quality randomized controlled trial; (4) Primary outcome measures included validated ADHD assessment tools such as Flanker, C-Span, Stroop, TMT-B, Stop-Signal Task, AWMAS, EF-Inhibition, ROCF, Stroop GNG, TGMD-2, Stroop C-W, WCST Total Correct, "IC", "WM", "CF", "SQ", CFT, and TAC; (5) Studies reported detailed statistical data including sample sizes, means, and standard deviations of assessment scale scores for both experimental and control groups.

2.2.2. Exclusion criteria

Studies were excluded if they: (1) Did not involve children aged 4-18 years with ADHD; (2) Lacked physical exercise interventions; (3) Included participants with severe physical or mental illness history; (4) Were not randomized controlled trials; (5) Did not assess ADHD symptoms as outcome measures; (6) Provided incomplete or unanalyzable data; (7) Contained duplicate published data; (8) Were case reports, conference abstracts, reviews, or other non-original research articles.

2.3. Statistical analysis

Meta-analysis was performed using Stata 14.0 software. Standardized mean difference (SMD) with 95% confidence intervals (95% CI) was used as the effect size measure. When studies reported multiple outcome measures, each effect size was included separately to preserve information across distinct outcome domains, consistent with approaches used in prior meta-analyses in this field.[10,12] Based on heterogeneity test results (I^2 statistic), a fixed-effects model was applied when $I^2 < 50\%$ and $P > 0.05$, indicating no significant heterogeneity among studies. A random-effects model was used when $I^2 \geq 50\%$ and $P \leq 0.05$, indicating significant heterogeneity.[22] Subgroup analyses were pre-specified in the PROSPERO protocol (CRD420251013415), with categorization cut-points based on prior literature and clinical guidelines for pediatric populations.

3. Results

3.1. Basic characteristics of included studies

Twenty publications were included, comprising 431 participants in experimental groups and 410 in control groups. Participants were distributed as follows: 121 in aerobic and endurance training groups, 99 in therapeutic and alternative exercise modality groups, 39 in mind-body and integrated health intervention groups, and 172 in skill-oriented and competitive sports training groups. The mean age of participants was 10.1 years. The majority of studies included school-age children (6–12 years). Gender information, where reported, indicated a higher proportion of male participants, consistent with ADHD epidemiology. Most studies did not specify ADHD subtype (predominantly inattentive, hyperactive-impulsive, or combined presentation), precluding subtype-specific analyses. ADHD outcome measures included Flanker, C-Span, Stroop, TMT-B, Stop-Signal Task, AWMAS, EF-Inhibition, ROCF, Stroop GNG, TGMD-2, Stroop C-W, WCST Total Correct, "IC", "WM", "CF", "SQ", CFT, and TAC. Table 1 presents details on

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Table 1
Features of the literature included in the study.

First Author (Year)	Sample Size (T/C)	Region	Age	Intervention Content	Intervention Duration,wk	Intervention Frequency	Length of Intervention Session, min	Outcomes
Ahmed(2011a)[23]	42/42	Egypt	11-16	Aerobic and Endurance Training	Short-term Intervention	High-frequency Intervention	Medium-duration Intervention	Behavior Rating scale (Attention)
Ahmed(2011b)[23]	42/42	Egypt	11-16	Aerobic and Endurance Training	Short-term Intervention	High-frequency Intervention	Medium-duration Intervention	Behavior Rating scale(Motor skills)
Ahmed(2011c)[23]	42/42	Egypt	11-16	Aerobic and Endurance Training	Short-term Intervention	High-frequency Intervention	Medium-duration Intervention	Behavior Rating scale(Task orientation)
Ahmed(2011d)[23]	42/42	Egypt	11-16	Aerobic and Endurance Training	Short-term Intervention	High-frequency Intervention	Medium-duration Intervention	Behavior Rating scale (Emotional &Oppositional behavior)
Ahmed(2011e)[23]	42/42	Egypt	11-16	Aerobic and Endurance Training	Short-term Intervention	High-frequency Intervention	Medium-duration Intervention	Behavior Rating scale (Academic& classroom behavior)
Benzing(2019a)[24]	28/23	Switzerland	8-12	Therapeutic and Alternative Exercise Modalities	Short-term Intervention	High-frequency Intervention	Medium-duration Intervention	Flanker task
Benzing(2019b)[24]	28/23	Switzerland	8-12	Therapeutic and Alternative Exercise Modalities	Short-term Intervention	High-frequency Intervention	Medium-duration Intervention	Color span backward
Bigelow(2021a)[25]	16/16	Canada	10-14	Mind-Body and Integrative Health Interventions	NA	Low-frequency Intervention	Short-duration Intervention	Stroop test
Bigelow(2021b)[25]	16/16	Canada	10-14	Mind-Body and Integrative Health Interventions	NA	Low-frequency Intervention	Short-duration Intervention	Leiter-3
Bigelow(2021c)[25]	16/16	Canada	10-14	Mind-Body and Integrative Health Interventions	NA	Low-frequency Intervention	Short-duration Intervention	TMT-B
Bustamante (2016a)[15]	19/16	USA	6-12	Aerobic and Endurance Training	Short-term Intervention	High-frequency Intervention	Medium-duration Intervention	Stop-signal inhibition task
Bustamante (2016b)[15]	19/16	USA	6-12	Aerobic and Endurance Training	Short-term Intervention	High-frequency Intervention	Medium-duration Intervention	Automated working memory assessment system
Chang(2014)[26]	14/13	China	5-10	Therapeutic and Alternative Exercise Modalities	Short-term Intervention	Mid-frequency Intervention	Very Long-duration Intervention	Go/Nogo Task, Motor ability (BMAT)
García-Gómez (2016a)[27]	9/5	Spain	7-14	Therapeutic and Alternative Exercise Modalities	Mid-term Intervention	Mid-frequency Intervention	Medium-duration Intervention	BASC(Hyperactivity)
García-Gómez (2016b)[27]	9/5	Spain	7-14	Therapeutic and Alternative Exercise Modalities	Mid-term Intervention	Mid-frequency Intervention	Medium-duration Intervention	BASC(Attention problems)
Gawrilow (2013)[28]	47/47	Germany	8.3-13.6	Skill-Oriented and Competitive Sports Training	NA	Low-frequency Intervention	Short-duration Intervention	Executive functioning (Inhibition task)
Halperin (2019a)[29]	26/26	USA	4-5	Skill-Oriented and Competitive Sports Training	Short-term Intervention	Low-frequency Intervention	Very Long-duration Intervention	ADHD Severity - KSADS
Halperin (2019b)[29]	26/26	USA	4-5	Skill-Oriented and Competitive Sports Training	Short-term Intervention	Low-frequency Intervention	Very Long-duration Intervention	ADHD Severity-P
Halperin (2019c)[29]	26/26	USA	4-5	Skill-Oriented and Competitive Sports Training	Short-term Intervention	Low-frequency Intervention	Very Long-duration Intervention	ADHD Severity–T
Halperin (2019d)[29]	26/26	USA	4-5	Skill-Oriented and Competitive Sports Training	Short-term Intervention	Low-frequency Intervention	Very Long-duration Intervention	ADHD Severity - CGI Severity
Hattabi(2019)[30]	20/20	Tunisia	9-12	Aerobic and Endurance Training	Mid-term Intervention	High-frequency Intervention	Very Long-duration Intervention	Cognitive functions (ROCF, Stroop test, Hayling test)
Jensen(2004a)[31]	11/8	Australia	8-13	Mind-Body and Integrative Health Interventions	Mid-term Intervention	Low-frequency Intervention	Medium-duration Intervention	Conners Parent Rating Scales
Jensen(2004b)[31]	11/8	Australia	8-13	Mind-Body and Integrative Health Interventions	Mid-term Intervention	Low-frequency Intervention	Medium-duration Intervention	Conners' Teacher Rating Scales
Kadri(2019a)[32]	20/20	Tunisia	14.5 ± 3.5	Skill-Oriented and Competitive Sports Training	Long-term Intervention	Mid-frequency Intervention	Medium-duration Intervention	Stroop test

(*continued on next page*)

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Table 1 (*continued*)

First Author (Year)	Sample Size (T/C)	Region	Age	Intervention Content	Intervention Duration,wk	Intervention Frequency	Length of Intervention Session, min	Outcomes
Kadri(2019b)[32]	20/20	Tunisia	14.5 ± 3.5	Skill-Oriented and Competitive Sports Training	Long-term Intervention	Mid-frequency Intervention	Medium-duration Intervention	Nowicki-Strickland Scale
Lufi(2011a)[33]	15/17	Israel	8.0-13.5	Skill-Oriented and Competitive Sports Training	Mid-term Intervention	Low-frequency Intervention	Very Long-duration Intervention	ASQ-P, YSR, CBCL
Lufi(2011b)[33]	17/15	Israel	8.0-13.5	Skill-Oriented and Competitive Sports Training	Mid-term Intervention	Low-frequency Intervention	Very Long-duration Intervention	ASQ-P, YSR, CBCL
Memarmoghaddam (2016)[14]	19/17	Iran	7-11	Skill-Oriented and Competitive Sports Training	Short-term Intervention	High-frequency Intervention	Very Long-duration Intervention	Stroop and Go-No-Go tests
Oh(2018a)[34]	17/15	Korea	6–12	Therapeutic and Alternative Exercise Modalities	Mid-term Intervention	Mid-frequency Intervention	Medium-duration Intervention	(ARS total score),(CBC), (SES), (PedsQL), (DCDQ), (CGI-S)
Oh(2018b)[34]	17/15	Korea	6–12	Therapeutic and Alternative Exercise Modalities	Mid-term Intervention	Mid-frequency Intervention	Medium-duration Intervention	(ARS inattention),(CBC), (SES), (PedsQL), (DCDQ), (CGI-S)
Oh(2018c)[34]	17/15	Korea	6–12	Therapeutic and Alternative Exercise Modalities	Mid-term Intervention	Mid-frequency Intervention	Medium-duration Intervention	(ARS hyperactivity-impulsivity),(CBC),(SES), (PedsQL), (DCDQ), (CGI-S)
Pan(2016a)[35]	16/16	china	6–12	Therapeutic and Alternative Exercise Modalities	Mid-term Intervention	Mid-frequency Intervention	Long-duration Intervention	Stroop test(baseline)
Pan(2016b)[35]	16/16	china	6–12	Therapeutic and Alternative Exercise Modalities	Mid-term Intervention	Mid-frequency Intervention	Long-duration Intervention	Stroop test(12 weeks from the baseline)
Pan(2016c)[35]	16/16	china	6–12	Therapeutic and Alternative Exercise Modalities	Mid-term Intervention	Mid-frequency Intervention	Long-duration Intervention	Stroop test(24weeks from the baseline)
Pan(2019a)[36]	15/15	china	7–12	Therapeutic and Alternative Exercise Modalities	Mid-term Intervention	Mid-frequency Intervention	Long-duration Intervention	TGMD-2, Stroop, WCST
Pan(2019b)[36]	15/15	china	7–12	Therapeutic and Alternative Exercise Modalities	Mid-term Intervention	Mid-frequency Intervention	Long-duration Intervention	TGMD-2, Stroop, WCST
Pan(2019c)[36]	15/15	china	7–12	Therapeutic and Alternative Exercise Modalities	Mid-term Intervention	Mid-frequency Intervention	Long-duration Intervention	Stroop Color–Word
Pan(2019d)[36]	15/15	china	7–12	Therapeutic and Alternative Exercise Modalities	Mid-term Intervention	Mid-frequency Intervention	Long-duration Intervention	WCST Total correct
Porter(1984a)[37]	12/11	USA	6-8	Mind-Body and Integrative Health Interventions	Short-term Intervention	Low-frequency Intervention	NA	Attention to task, Impulsivity, Locus of Control
Porter(1984b)[37]	12/11	USA	6-8	Mind-Body and Integrative Health Interventions	Short-term Intervention	Low-frequency Intervention	NA	Impulsivity
Porter(1984c)[37]	12/11	USA	6-8	Mind-Body and Integrative Health Interventions	Short-term Intervention	Low-frequency Intervention	NA	Locus of Control
Qiu(2023)[38]	40/40	china	6-12	Aerobic and Endurance Training	Mid-term Intervention	High-frequency Intervention	Medium-duration Intervention	Inhibitory control, working memory, cognitive flexibility, sleep quality
Da Silva(2019a)[39]	10/10	Brazil	11-14	Skill-Oriented and Competitive Sports Training	Short-term Intervention	Mid-frequency Intervention	Medium-duration Intervention	Child Depression Inventory (CDI)
Da Silva(2019b)[39]	10/10	Brazil	11-14	Skill-Oriented and Competitive Sports Training	Short-term Intervention	Mid-frequency Intervention	Medium-duration Intervention	Perceived Stress Scale (PSS-14)
Da Silva(2019c)[39]	10/10	Brazil	11-14	Skill-Oriented and Competitive Sports Training	Short-term Intervention	Mid-frequency Intervention	Medium-duration Intervention	Cognitive Flexibility (Test of Trails)
Da Silva(2019d)[39]	10/10	Brazil	11-14	Skill-Oriented and Competitive Sports Training	Short-term Intervention	Mid-frequency Intervention	Medium-duration Intervention	Selective Attention (TAC Cancellation Attention Test)
Da Silva(2019e)[39]	10/10	Brazil	11-14	Skill-Oriented and Competitive Sports Training	Short-term Intervention	Mid-frequency Intervention	Medium-duration Intervention	Motor Coordination
Vaziri(2014)[40]	18/18	Iran	NA	Skill-Oriented and Competitive Sports Training	NA	NA	NA	Conners Parent Rating Scale (CPRS-48)

Note:(Short-term Intervention) The intervention period is less than 12 weeks; (Mid-term Intervention) The intervention period was 12-24 weeks; (Long-term Intervention) Intervention period of more than 24 weeks.

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(Low-frequency Intervention) once a week; (Mid-frequency Intervention) twice a week; (High-frequency Intervention) 3 to 5 times or more per week.
(Short-duration Intervention) Each intervention lasts less than 30 min; (Medium-duration Intervention) Each intervention lasts 30-60 min; (Long-duration Intervention) Each intervention lasts 60 to 90 min.

exercise intervention cycles, intervention frequency, and single intervention duration.

3.1.1. Literature screening and data extraction

Relevant literature was identified from PubMed, Web of Science, Cochrane Library, ScienceDirect, and EBSCO databases using predetermined search strategies. References were imported into EndNote with duplicates removed. Two researchers independently screened titles and abstracts based on inclusion and exclusion criteria, then conducted full-text assessments to determine final included randomized controlled trials following the PRISMA flow diagram (Fig. 2). Data were independently extracted using a self-designed form including author, publication date, study design, sample characteristics, interventions, and outcome measures. Original data were obtained from authors when statistical data was presented only in graphs. All data were compiled using Excel software with disagreements resolved through discussion or arbitration by a third researcher to ensure objective and reliable results throughout the double-blind screening process.

3.1.2. Quality assessment of literature

Two researchers independently assessed study quality using Review Manager 5.3 software based on Cochrane Handbook criteria. Assessment included: (1) Random sequence generation; (2) Allocation concealment; (3) Blinding of participants and personnel; (4) Blinding of outcome assessment; (5) Incomplete outcome data; (6) Selective reporting; (7) Other bias. Each item was rated as low risk, unclear risk, or high risk based on study descriptions and methodological adequacy. Disagreements were resolved through discussion. Quality assessment results are summarized in risk of bias graphs (Figs. 3 and 4), displaying bias conditions and distribution across studies.

3.2. Heterogeneity test

Statistical threshold for heterogeneity assessment was preset at $I^2 > 50\%$. Using the I^2 statistic to test inter-study heterogeneity, $I^2 = 86.8\%$ and $P = 0.00$ were found, indicating significant statistical heterogeneity among included studies ($I^2 > 50\%$, $P < 0.05$) (Fig. 5). This high heterogeneity likely stems from: (1) differences in sample characteristics, such as age, gender, ADHD subtype, and disease severity; (2) variations in intervention measures, including exercise type, frequency, duration, and intensity; and (3) differences in outcome measurements, such as diverse assessment tools and evaluation time points.[39] According to heterogeneity magnitude criteria, an I^2 value of 86.8% indicates substantial heterogeneity among studies.

3.3. Sensitivity analysis

Sensitivity analysis was conducted to verify the stability and reliability of results. Using a leave-one-out method, the pooled effect size was recalculated after excluding each study in turn. Results showed that exclusion of any single study had minimal impact on the overall outcome, with little change in the pooled effect size and confidence intervals (Fig. 6). This confirms the reliability of the conclusion that physical exercise intervention improves symptoms in children with ADHD.

3.4. Publication bias analysis

Egger's test was used to assess potential publication bias in the included studies (Fig. 7). Results showed a P-value of 0.33 (>0.05) for Egger's test, indicating no significant publication bias in the meta-analysis of physical exercise interventions for children with ADHD. Additionally, the funnel plot displayed a relatively symmetrical

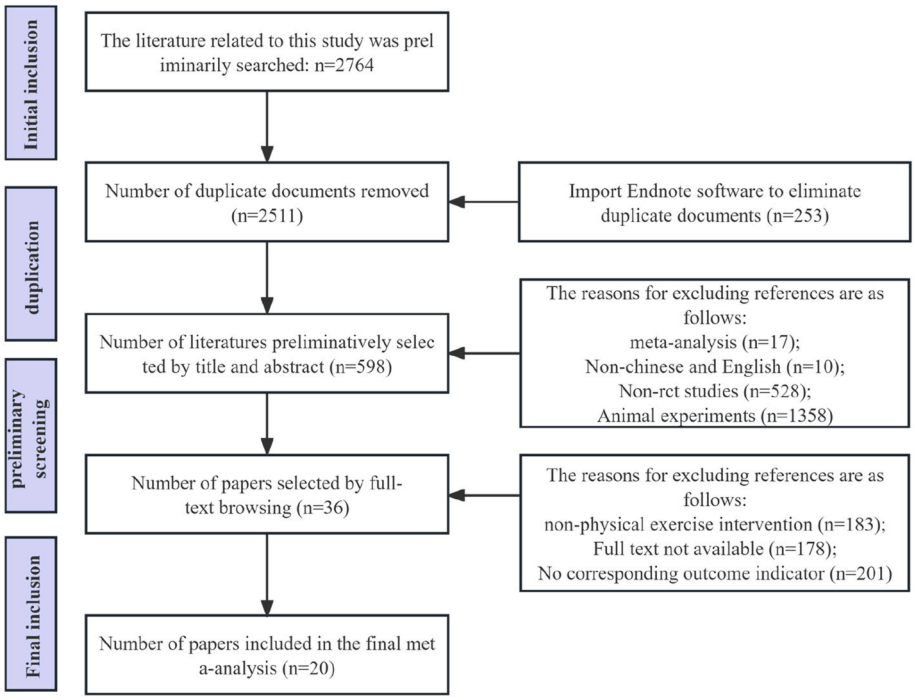


Fig. 2. Literature screening flow chart.
*The number of articles retrieved from each database was as follows:PubMed(n = 91), Web of Science(n = 610). Cochrane Library(n = 205), ScienceDirect(n = 1763), EBSCO(n = 95).

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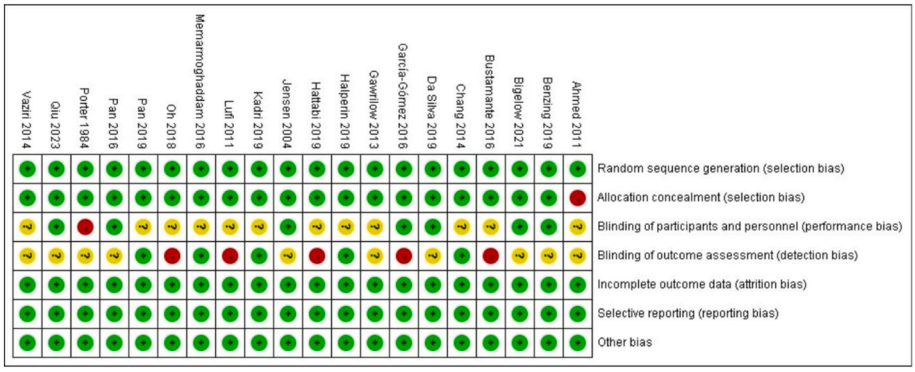


Fig. 3. Assessment of the risk of bias for included studies. +, low bias risk; -, high bias risk; ? unclear bias risk.

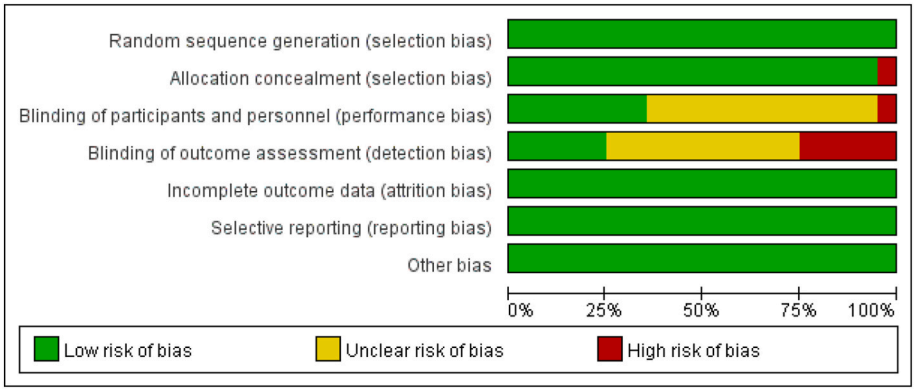


Fig. 4. Distribution of studies across the bias ranking for each type of bias.

distribution of studies, with data points generally balanced on both sides.

3.5. Meta-regression analysis

A random-effects meta-regression model was employed to assess the impact of different exercise prescription parameters on effect sizes, with standardized mean difference (SMD) as the dependent variable and intervention type, intervention duration, intervention frequency, single session duration, and outcome measures as covariates. Results demonstrated that none of the covariates reached statistical significance (all $P > 0.05$): intervention type ($P = 0.37$), intervention duration ($P = 0.88$), intervention frequency ($P = 0.78$), single session duration ($P = 0.53$), and outcome measures ($P = 0.18$). These findings indicate that no single variable emerged as a statistically significant predictor of effect size variation, suggesting that the substantial between-study heterogeneity ($I^2 = 86.8\%$) likely stems from complex multifactorial interactions rather than individual parameter effects. Given that meta-regression analysis failed to identify significant predictors, pre-specified subgroup analyses were conducted to compare the differential effects of various exercise parameters on ADHD symptoms in children (Table 2).

3.6. Subgroup analysis

Subgroup analyses were conducted based on intervention component, duration, frequency, single session length, and outcome measures to explore the impact of different exercise prescription parameters. In addition, when conducting subgroup analyses, subgroups with effect sizes of less than five should be interpreted with caution. Results showed that all five subgroups significantly improved ADHD symptoms compared to control groups (Table 3). Therapeutic and alternative exercise modalities demonstrated greater effectiveness (SMD = 0.45, 95% CI [0.26-0.65]).

Regarding intervention duration, samples were divided into short (<12 weeks), medium (12-24 weeks), and long (>24 weeks) periods. Medium-duration interventions showed significant positive effects (SMD = 0.28, 95% CI [0.11-0.46], $P < 0.01$), while long-duration interventions showed no significant effect ($P = 0.42$). For intervention frequency, samples were categorized into low (1/week), medium (2/week), and high (3-5/week) frequency groups. Medium-frequency interventions demonstrated superior effects (SMD = 0.52, 95% CI [0.33-0.71]). Single session lengths were divided into short (<30 min), medium (30-60 min), long (60-90 min), and very long (>90 min). Long-duration sessions (60-90 min) outperformed other groups (SMD = 1.01, 95% CI [0.71-1.31]). Outcome measures were categorized into behavioral assessment, cognitive function assessment, ADHD severity assessment, and emotional and social behavior assessment groups. Cognitive function assessment showed the largest effect sizes (SMD = 0.45, 95% CI [0.30-0.60], $P < 0.01$).

Subgroup analysis revealed that therapeutic and alternative exercise modalities, medium-duration (12-24 weeks) interventions, medium-frequency (2/week) interventions, and long-duration (60-90 min) sessions demonstrated superior effectiveness. Cognitive function assessment scales appeared most sensitive for detecting exercise intervention benefits in ADHD children.

4. Discussion

This meta-analysis systematically evaluated the impact of physical exercise interventions on core symptoms of ADHD in children. Key findings include: (1) Physical exercise interventions may improve core ADHD symptoms such as attention deficit and hyperactivity-impulsivity; (2) Exploratory subgroup analyses suggest that different exercise prescription parameters may be associated with varying intervention

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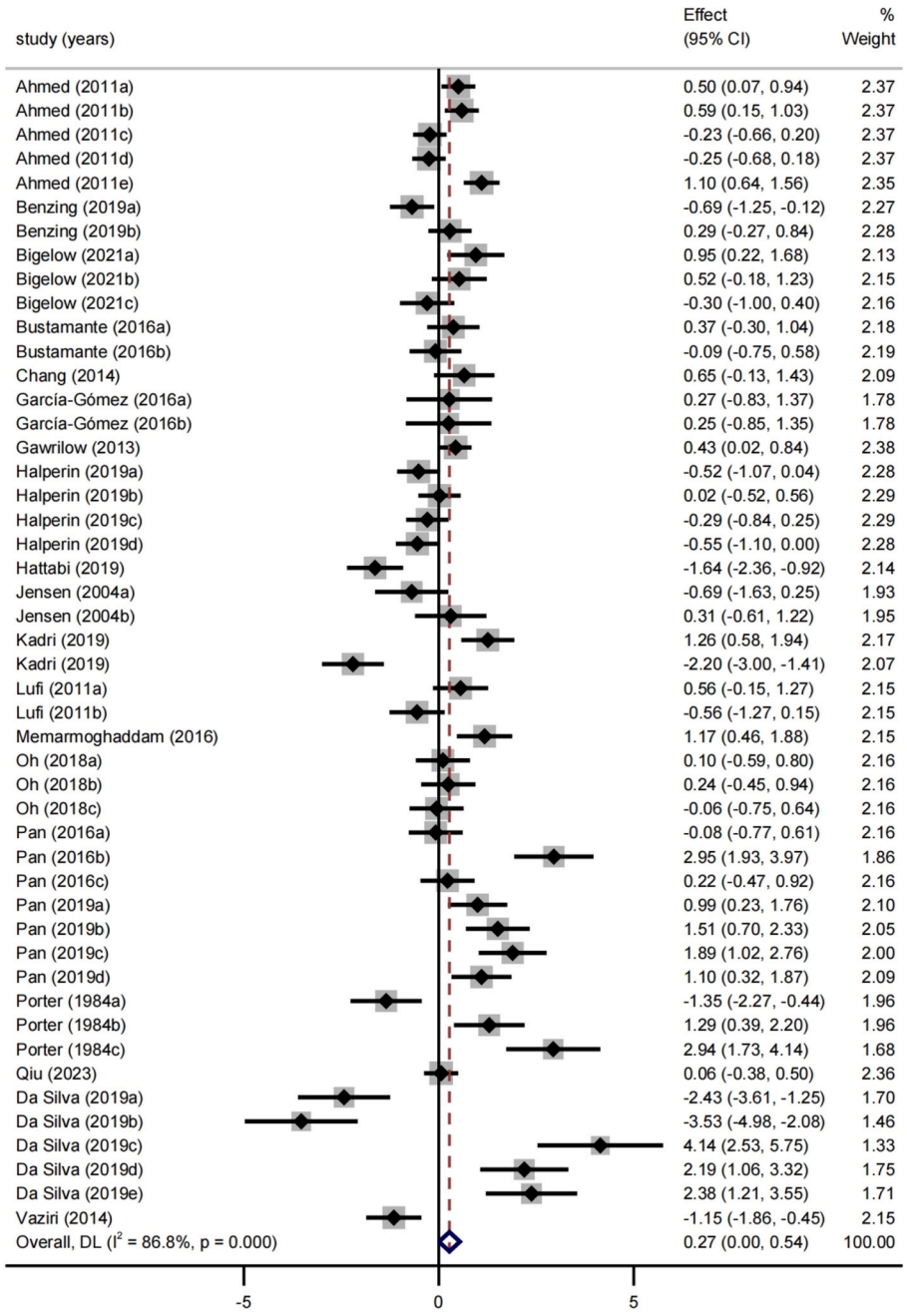


Fig. 5. Heterogeneity analysis of physical exercise on ADHD children.

effectiveness, with therapeutic and alternative exercise programs conducted twice weekly for 60-90 min per session being most effective; (3) Intervention duration of 12-24 weeks appears to be associated with greater therapeutic effects. This study provides high-quality evidence supporting physical exercise as a potential adjunctive approach for ADHD and offers preliminary references for developing evidence-based exercise prescriptions. These findings are broadly consistent with prior meta-analyses by Cerrillo-Urbina et al.,[10] Vysniauske et al.,[11] and Liang et al.[12] in confirming the positive effects of exercise on ADHD-related outcomes. The present study extends this literature by simultaneously evaluating multiple prescription dimensions, although the lack of significant moderators in meta-regression indicates that these prescription-level findings should be regarded as hypothesis-generating. The following subsections discuss these findings in detail.

4.1. *Exercise modality effectiveness*

Therapeutic and alternative exercise modalities demonstrated greater effectiveness than aerobic and endurance training (SMD = 0.45, 95% CI [0.26-0.65]). These modalities may provide more structured and cognitively demanding activities that challenge executive functions. Chan et al.[41] similarly reported that therapeutic and alternative exercises are more beneficial for treating ADHD children. Cerrillo-Urbina et al.[42] demonstrated through meta-analysis that therapeutic and alternative exercise methods, such as yoga and martial arts, provide structured and effective interventions for children with ADHD, showing significant improvements in core symptoms compared to conventional physical activities. Hoza et al.[43] propose that the superiority of therapeutic and alternative exercise modalities may stem from their multifaceted challenges to executive functions, including inhibitory control,

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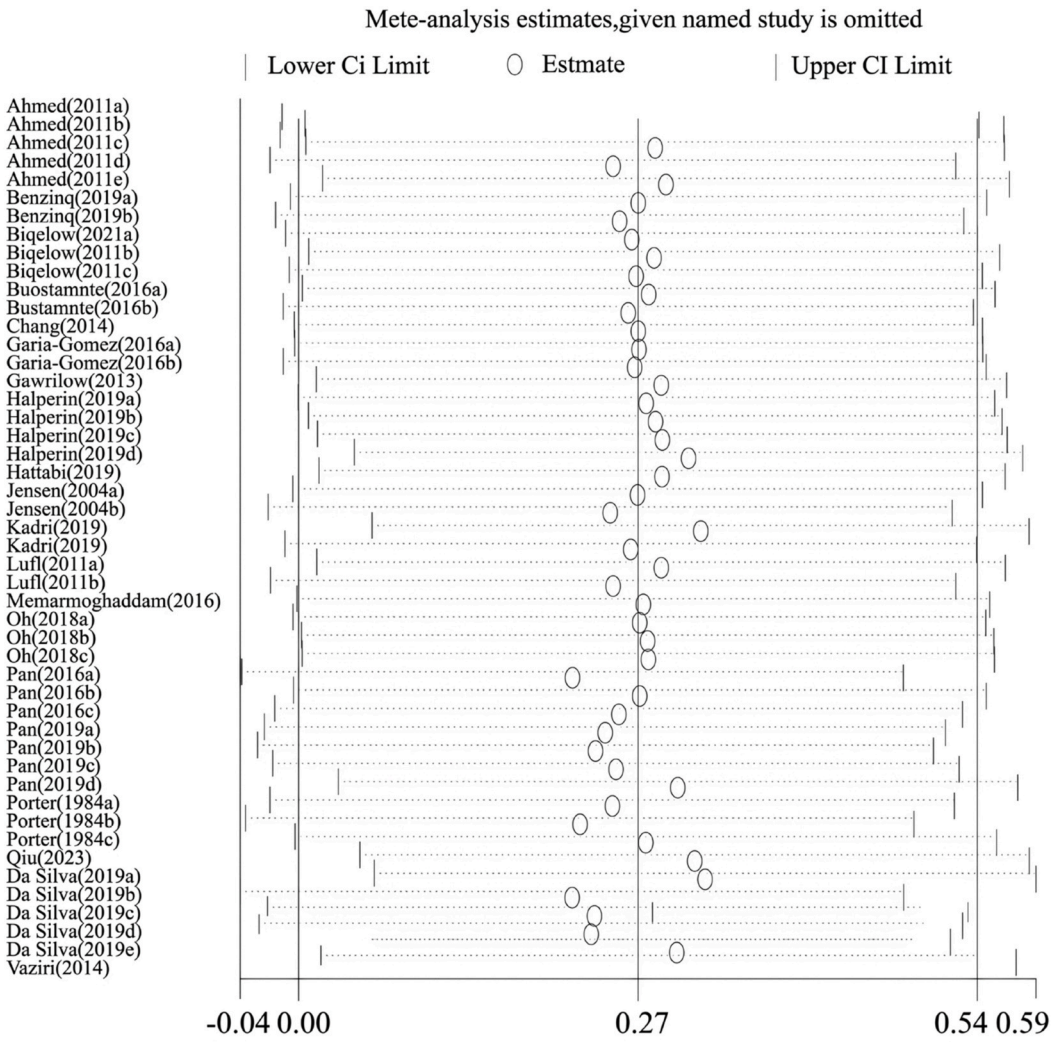


Fig. 6. Sensitivity analysis.

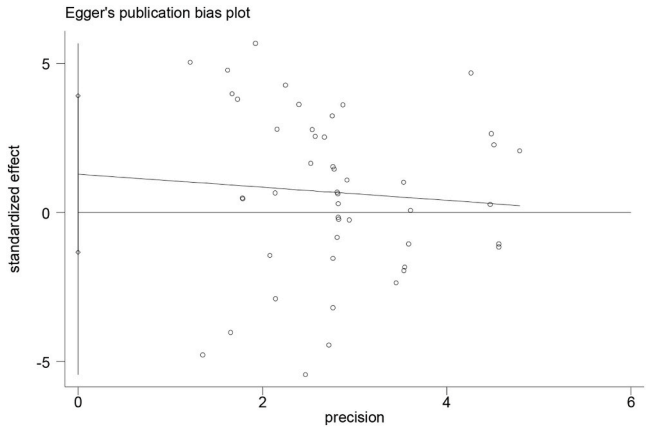


Fig. 7. Egger method tests publication bias.

working memory, and cognitive flexibility. Neuroimaging studies by Chaddock-Heyman et al.[44] show that regular physical exercise can increase gray matter volume in children's prefrontal cortex, an area closely related to executive function.

Table 2
Meta-regression analysis of the intervention effect of physical exercise on ADHD children.

_ES	Coef.	Std. Err.	t	$P > t$	[95% Conf.	Interval]
Intervention content	0.43	0.47	0.91	0.37	−0.52	1.38
Intervention cycle	0.92	0.59	0.16	0.88	−1.09	1.28
Intervention frequency	0.10	0.37	0.28	0.78	−0.64	0.84
length of each intervention session	0.40	0.63	0.63	0.53	−0.87	1.67
outcome	1.10	0.76	1.44	0.18	−0.55	2.74

4.2. Duration and frequency parameters

Subgroup analysis revealed that medium-term interventions (12-24 weeks) were more effective than short-term (<12 weeks) or long-term (>24 weeks) interventions ($P < 0.01$). These findings align with Liang et al.,[45] demonstrating that 12-week exercise programs effectively enhance children's compliance and engagement. Medium-frequency interventions (twice weekly) demonstrated superior effects (SMD = 0.52, 95% CI [0.33-0.71]), possibly reflecting a balance between sufficient exercise stimulus and adequate recovery time. García-Gómez et al.[27] propose that physical exercise may improve ADHD symptoms by regulating neurotransmitter systems, particularly the dopamine system.[46] Wigal et al.[47] found that acute exercise increases dopamine levels

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Table 3
Subgroup analysis of the intervention effect of physical exercise on ADHD children.

group		number	SMD	95% CI		*P*	I^2/%	The p-value of SMD
Intervention Content	Aerobic and Endurance Training	9	0.16	0.00	0.32	0.00	85.10	0.05
	Therapeutic and Alternative Exercise Modalities	15	0.45	0.26	0.65	0.00	79.10	0.00
	Mind-Body and Integrative Health Interventions	8	0.33	0.03	0.63	0.00	84.80	0.03
	Skill-Oriented and Competitive Sports Training	16	−0.03	−0.20	0.14	0.00	91.40	0.74
Intervention cycle	Short-term Intervention	23	0.17	0.04	0.30	0.00	88.60	0.01
	Mid-term Intervention	18	0.28	0.11	0.46	0.00	82.40	0.00
	Long-term Intervention	2	−0.21	−0.73	0.31	0.00	97.60	0.42
Intervention Frequency	Mid-frequency Intervention	20	0.52	0.33	0.71	0.00	90.00	0.00
	Low-frequency Intervention	15	0.06	−0.11	0.23	0.00	79.30	0.52
	High-frequency Intervention	12	0.16	0.01	0.30	0.00	84.40	0.04
Length of Each Intervention	Medium-duration Intervention	24	0.16	0.03	0.29	0.00	87.20	0.02
	Short-duration Intervention	4	0.40	0.11	0.69	0.10	51.30	0.01
	Very Long-duration Intervention	9	−0.18	−0.39	0.03	0.00	81.40	0.09
	Long-duration Intervention	7	1.01	0.71	1.31	0.00	82.20	0.00
outcome	Behavioral Assessment Group	10	0.23	0.06	0.40	0.00	81.00	0.01
	Cognitive Function Assessment Group	20	0.45	0.30	0.60	0.00	87.00	0.00
	Emotional and Social Behavior Assessment Group	11	−0.23	−0.51	0.06	0.00	92.10	0.13
	ADHD Severity Assessment Group	7	−0.20	−0.42	0.03	0.43	0.00	0.09

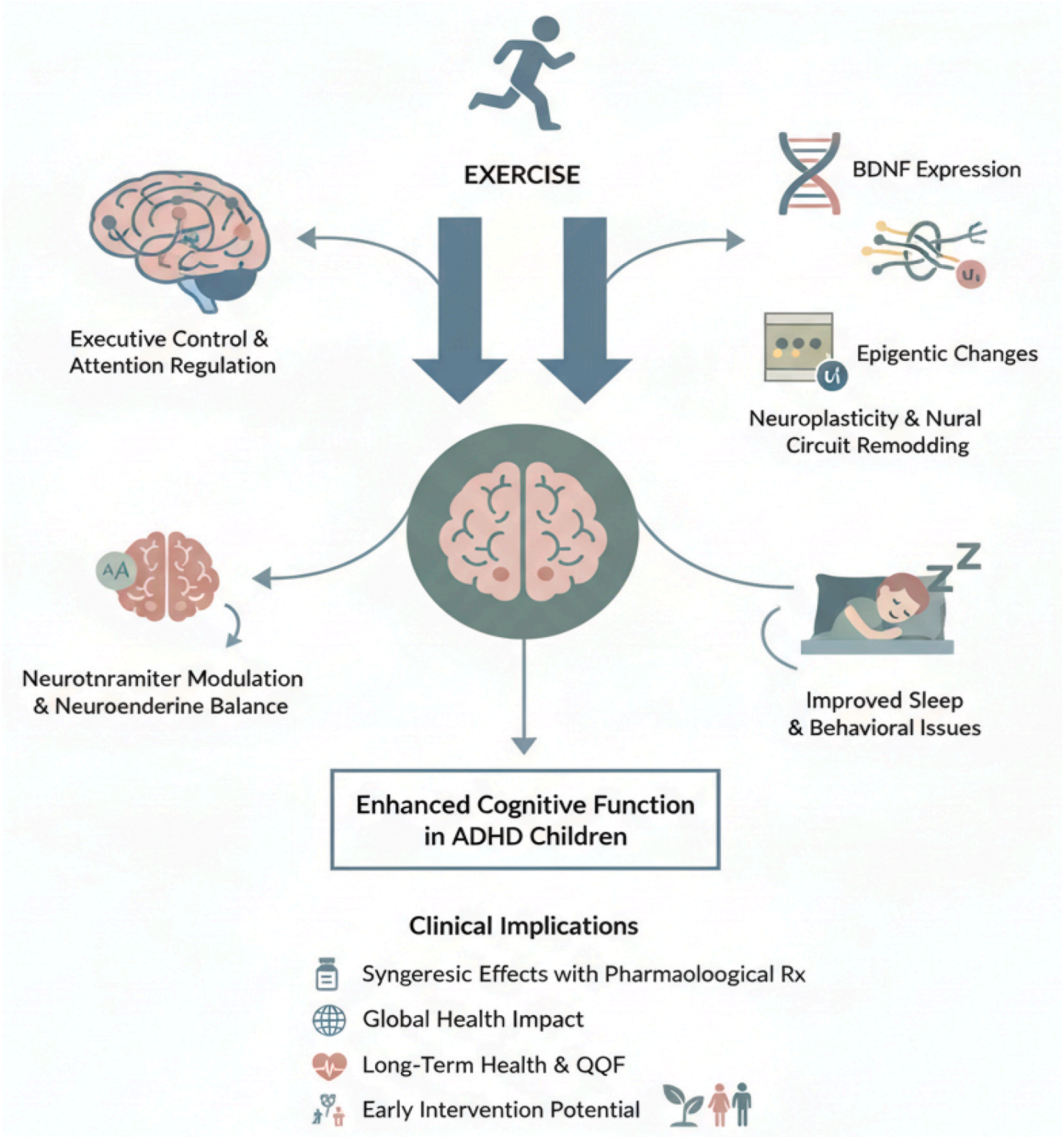


Fig. 8. Proposed neurobiological mechanisms underlying exercise-induced improvements in ADHD symptoms.

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in ADHD children's brains, directly related to improved attention and executive function.

Long exercise sessions (60-90 min) demonstrated superior efficacy (SMD = 1.01, 95% CI [0.71-1.31]) compared to shorter or longer sessions, suggesting a non-linear, inverted U-shaped dose-response relationship. This duration may allow sufficient time for structured exercise components to engage sustained attention and executive control processes. Christiansen et al.[48] demonstrated that structured 60-min exercise sessions significantly elevated serum BDNF levels in children with ADHD, correlating with improved attentional performance. Verret et al.[49] found that 60-90 min physical activity sessions improved cognitive inhibition and working memory through enhanced cerebral blood flow.

4.3. Underlying neurobiological mechanisms

Cognitive function assessments (SMD = 0.45, 95% CI [0.30-0.60]) best evaluated effects on ADHD children, suggesting exercise particularly enhances prefrontal-striatal circuit function. This circuit plays a crucial role in executive control and attention regulation.[50] Exercise may promote cognitive improvement through multiple pathways, including enhanced cerebral blood flow and neurotransmitter modulation,[51] although the specific mechanisms in children with ADHD remain to be elucidated. Grassmann et al.[52] showed that exercise can improve sleep quality in ADHD children, which correlates with improvements in attention and behavioral issues.

These findings have significant clinical implications. Chang et al.[26] found that complex exercise patterns may have greater positive impact on executive functions in ADHD children. Jensen and Kenny et al.[31] show that combining physical activities with pharmacological treatment may produce synergistic effects, potentially improving treatment efficacy while allowing reduced drug dosages. Faraone et al.[53] note that physical exercise may play a crucial role in addressing ADHD as a global health issue, particularly in areas with limited medical resources. Smith et al.[54] suggest that long-term adherence to physical exercise may improve not only core ADHD symptoms but also patients' overall health and quality of life. Halperin et al.[55] research suggests that early exercise interventions may help prevent or mitigate the development of ADHD symptoms, indicating potential applications for at-risk children (Fig. 8).

4.4. Conclusion

This study provides high-quality evidence supporting physical exercise as an adjunctive treatment for ADHD, with significant implications for clinical practice. Based on exploratory subgroup analyses, clinicians and parents may consider therapeutic and alternative exercises twice weekly for 60-90 min per session when developing exercise plans for ADHD children, although meta-regression did not identify statistically significant predictors of effect size variation. Future research should further explore the mechanisms of exercise interventions, optimize exercise prescription development and implementation processes, and evaluate synergistic effects with other treatments (e.g., medication, behavioral therapy) to provide more comprehensive theoretical and practical guidance for evidence-based, multimodal ADHD treatment.

4.5. Limitations and future research directions

This study incorporated numerous high-quality RCTs, providing robust evidence for the positive impact of physical exercise on children with ADHD. However, several limitations warrant consideration. First, the included studies primarily sampled from developed countries in Europe and America, potentially affecting result representativeness. Second, although all included studies were RCTs, they varied in intervention quality control measures, such as supervision and exercise intensity measurement. Third, most trials had relatively short follow-up periods (typically <6 months), lacking examination of long-term exercise intervention effects. Fourth, the substantial heterogeneity (I^2 = 86.8%) limits the interpretability of the pooled effect size, and none of the tested moderators in meta-regression significantly explained this variability (all $P > 0.05$), suggesting that unmeasured factors may also contribute. Fifth, including multiple effect sizes from the same study may underestimate standard errors; future research should consider multilevel models or robust variance estimation to address this limitation. Sixth, blinding of outcome assessors was not consistently implemented across included studies, which may have contributed to inflated effect sizes.

To address these limitations, future studies should conduct long-term (>2 years) randomized controlled trials to evaluate sustained impacts of physical exercise on ADHD symptoms. Understanding underlying mechanisms through functional magnetic resonance imaging would provide insights into how different exercise types affect brain functional connectivity in ADHD children. Research should also develop individualized exercise prescriptions considering ADHD subtypes, comorbid conditions, and personal preferences, while exploring optimal combinations of physical exercise and pharmacological treatments. Expanding research to diverse geographic and cultural contexts will be essential for assessing intervention effectiveness across different populations and resource settings. These efforts aim to establish evidence-based management strategies that improve long-term outcomes for children with ADHD.

Ethics statement

Not applicable; this study is a systematic review and meta-analysis of previously published studies. No new data were collected from human participants or animals; therefore, ethical approval was not required.

Availability of data and materials

All data generated or analyzed during this study are included in this published article and its supplementary information files.

Author contributions

B.C. conceptualized the study, designed the methodology, curated the data, performed formal analysis, and wrote the original draft. J.L. validated the findings, conducted investigations, provided resources, and reviewed and edited the manuscript. Z.S. supervised the project, contributed to conceptualization, and reviewed and edited the manuscript. All authors read and approved the final manuscript.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no competing interests.

Acknowledgments

The authors gratefully acknowledge all individuals who contributed to this study, including colleagues, reviewers, and support staff, whose collective efforts were invaluable to the completion of this research.

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